RESULTS — TYLER EXAM SET A PAPER 2

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20 attempted, 0 left blank · total time on paper 1:14:44

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TMUA SCORE

5.9 2024+ scale 7.0 2017–2023 scale

Estimated from our grade boundaries — not an official TMUA result.

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BY DIFFICULTY

4.4	1 / 1
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6.8	1 / 1
6.9	1 / 1
7.0	1 / 1
7.3	1 / 1
7.5	1 / 2

Per-question breakdown

#	YOUR ANSWER	CORRECT	RESULT	TIME		
1	C	C	✓	2:07	Bookmark	Review
2	C	C	✓	2:25	Bookmark	Review
3	C	B	✗	1:49	Bookmark	Review
4	B	B	✓	4:11	Bookmark	Review
5	B	E	✗	1:09	Bookmark	Review
6	D	B	✗	6:05	Bookmark	Review
7	B	B	✓	2:49	Bookmark	Review
8	D	D	✓	3:24	Bookmark	Review
9	C	C	✓	4:34	Bookmark	Review
10	A	C	✗	6:27	Bookmark	Review
11	E	E	✓	3:26	Bookmark	Review
12	E	E	✓	3:28	Bookmark	Review
13	A	A	✓	2:24	Bookmark	Review
14	A	A	✓	3:49	Bookmark	Review
15	D	D	✓	3:14	Bookmark	Review
16	D	A	✗	5:14	Bookmark	Review
17	G	G	✓	4:10	Bookmark	Review
18	B	D	✗	2:50	Bookmark	Review
19	D	D	✓	3:03	Bookmark	Review
20	C	C	✓	8:06	Bookmark	Review

3 C B ✕ 1:49 Bookmark Hide

QUESTION

Consider the following statements about the polynomial function $f(x)$:

P: The trapezium rule with n equal intervals is used to estimate $\int_a^b f(x)dx$ and produces an underestimate.

Q: $f''(x) < 0$ for $a \leq x \leq b$.

Which of the following is correct?

A P is sufficient but not necessary for Q.

B P is necessary but not sufficient for Q. CORRECT

C P is both necessary and sufficient for Q. YOUR CHOICE

D P is neither necessary nor sufficient for Q.

SOLUTION

If Q is true, then $f''(x) < 0$ throughout $[a, b]$, so the graph of f is concave down on the whole interval. Each straight-line chord used by the trapezium rule therefore lies below the graph, so the trapezium rule gives an underestimate. Hence $Q \Rightarrow P$, so P is **necessary** for Q.

However, P **does not imply** Q in general. For example, take $f(x) = x^2 - x^4$ on $[-1, 1]$ with $n = 1$. Since $f(-1) = f(1) = 0$, the trapezium rule gives 0, while the exact integral is

$$\int_{-1}^{1} (x^2 - x^4)\, dx = \frac{4}{15} > 0.$$

Thus P is true. However, $f''(x) = 2 - 12x^2$, so $f''(0) = 2 > 0$, making Q false.

Therefore P is necessary but not sufficient for Q.

Logic Deduction | Trapezium Rule | Difficulty 6.2

QUESTION

Consider the equation

$$x^2 - 2kx + k + 2 = 0$$

Let $n(k)$ represent the number of distinct real solutions to this equation.
Which of the following statements is **necessarily** true?

A $n(k) = 0$ for $k < -1$

B $n(k) = 0$ for $k = -1$ YOUR CHOICE

C $n(k) = 1$ for $k = 1$

D $n(k) = 2$ for $1 < k < 2$

E $n(k) = 2$ for $k < -2$ CORRECT

SOLUTION

The number of distinct real solutions is determined by the discriminant:

$$b^2 - 4ac = (-2k)^2 - 4(k+2) = 4(k^2 - k - 2) = 4(k-2)(k+1).$$

Hence:

Case 1: $n(k) = 2$ if and only if $k < -1$ or $k > 2$;

Case 2: $n(k) = 1$ if and only if $k = -1$ or $k = 2$;

Case 3: $n(k) = 0$ if and only if $-1 < k < 2$.

Therefore the only statement that can be deduced is $n(k) = 2$ for all $k < -2$ as this lies within the region $k < -1$.

Logic Deduction | General Number of Solutions | Difficulty 5.2

QUESTION

A student attempts to solve

$$\log_2(3x-1) = 1 + 2\log_2(x-1)$$

Their working is shown below. At each step, the equation written is intended to be equivalent to the equation in the preceding step, with the same solution set.

Step 1: $\log_2(3x-1) - 2\log_2(x-1) = 1$

Step 2: $\log_2(3x-1) - \log_2((x-1)^2) = 1$

Step 3: $\log_2 \dfrac{3x-1}{(x-1)^2} = 1$

Step 4: $\dfrac{3x-1}{(x-1)^2} = 2$

Step 5: $2x^2 - 7x + 3 = 0$

Step 6: $(2x-1)(x-3) = 0$

Step 7: $x = \frac{1}{2}$ or $x = 3$

Which of the following statements is true?

A The attempt is incorrect and the first mistake appears in Step 1

B The attempt is incorrect and the first mistake appears in Step 2 — CORRECT

C The attempt is incorrect and the first mistake appears in Step 3

D The attempt is incorrect and the first mistake appears in Step 4 — YOUR CHOICE

E The attempt is incorrect and the first mistake appears in Step 5

F The attempt is incorrect and the first mistake appears in Step 6

G The attempt is incorrect and the first mistake appears in Step 7

H The solution is correct.

QUESTION

A triangle ABC is to be drawn with the following measurements:

$$AB = 5a, \quad BC = \sqrt{17}a, \quad \text{and } \angle BAC = \theta^\circ, \text{ where } \tan\theta = \tfrac{4}{3}.$$

Given that there are two non-congruent possible triangles satisfying these conditions, what is the ratio of the area of the larger triangle to the area of the smaller triangle?

A $\sqrt{17} : 2$ YOUR CHOICE

B $5 : 2$

C $2 : 1$ CORRECT

D $\sqrt{17} : 5$

E $4 : 1$

SOLUTION

We can set $a = 1$, since changing a only scales the triangles and does not affect the ratio of their areas.

Since $\tan\theta = \frac{4}{3}$, we have $\cos\theta = \frac{3}{5}$.

Let $AC = x$. Using the cosine rule,

$$17 = 5^2 + x^2 - 2(5)(x)\frac{3}{5} \quad \Rightarrow \quad x^2 - 6x + 8 = 0,$$

giving $x = 4$ or $x = 2$. These correspond to the two possible triangles.

Taking AC as the base, the two triangles have the same height, since AB and θ are the same. Therefore their area ratio is the same as the ratio of their bases, which are the two lengths just found: 4 and 2.

Thus the required area ratio is $4 : 2 = 2 : 1$.

General Trigonometry Geometry Difficulty 6.6

QUESTION

Let x be a real number. Consider the following statements.

A: $x < 2$.

B: $0 < x < 1$.

C: $x^2 < 1$.

D: $|x - 1| < 2$.

E: $x^2 < 4$.

Which statement is a necessary condition for exactly three of the other four statements?

A A CORRECT

B B

C C

D D YOUR CHOICE

E E

SOLUTION

Statement A doesn't imply any other statement, since a negative number such as -5 is not contained in any of the other statements.

Statement B implies all the other statements as any number between 0 and 1 satisfies all the other conditions.

Statement C implies statements A, D, E

Statement D doesn't imply any other statement, since a number $2 < x < 3$ is not contained in any of the other statements.

Statement E implies A only

Thus A is necessary for B, C and E,
(and note that C is a sufficient condition for exactly 3 of the other four statements).

QUESTION

Consider the statement:

For every real number x, there exists a real number y such that for every integer n,

$$\sin(x + y + 2n\pi) = 1$$

Which of the following is the negation of this statement?

A For every real x, there exists a real y such that $\sin(x + y + 2n\pi) \neq 1$ for all integers n.

B For every real x and every real y there exists an integer n such that $\sin(x + y + 2n\pi) \neq 1$. YOUR CHOICE

C For every real x, there exists a real y such that $\sin(x + y + 2n\pi) \neq 1$ for some integer n.

D There exists a real x, such that for every real y there exists an integer n such that $\sin(x + y + 2n\pi) \neq 1$. CORRECT

E There exists a real x and a real y such that $\sin(x + y + 2n\pi) \neq 1$ for some integer n.

SOLUTION

Write the original statement symbolically as

$$\forall x \in \mathbb{R},\ \exists y \in \mathbb{R},\ \forall n \in \mathbb{Z}, \quad \sin(x + y + 2n\pi) = 1.$$

To negate a statement, reverse each quantifier and negate the final condition:

$$\forall \longleftrightarrow \exists, \qquad \exists \longleftrightarrow \forall.$$

Hence the negation is

$$\exists x \in \mathbb{R},\ \forall y \in \mathbb{R},\ \exists n \in \mathbb{Z} \ \text{ such that } \ \sin(x + y + 2n\pi) \neq 1.$$

In words: There exists a real x, such that for every real y there exists an integer n such that $\sin(x + y + 2n\pi) \neq 1$.

Logic Negation Difficulty 5.5